

Solutions to JEE Main - 7 | JEE-2022

PHYSICS

SECTION-1

- 1.(D) Time period of a simple pendulum, $T = 2\pi\sqrt{\frac{L}{g}}$

$$\text{So, } T_C = 2\pi\sqrt{\frac{L_C}{g}} \text{ and } T_H = 2\pi\sqrt{\frac{L_H}{g}}$$

$$\text{Also, } L_H = L_C(1 + \alpha(35 - 5)) \Rightarrow L_H = L_C(1 + 30\alpha)$$

$$\text{Therefore, } \frac{T_C}{T_H} = \left(\frac{L_C}{L_H}\right)^{1/2} = \frac{1}{(1 + 30\alpha)^{1/2}} \Rightarrow \frac{T_C}{T_H} = 1 - 15\alpha$$

- 2.(C) Work done in isothermal process,

$$W = nRT \log_e \left(\frac{V_2}{V_1}\right) \Rightarrow W = (1)(R)(300) \log_e(8) = (900 \log_e 2)R = 900(0.693)(8.31) = 5183 \text{ J}$$

- 3.(C) Let the equation of the SHM be:

$$x(t) = A \sin(2\pi ft)$$

So, we can see that the particle is at its mean position at $t=0$ and at $x = \frac{A}{2}$ for the first time at

$$t = \frac{1}{12f}$$

- 4.(C) Rate of radiation of heat from the surface of a block body is given by Stefan's law:

$$\frac{dQ}{dt} = \sigma AT^4$$

Here, σ is the Stefan-Boltzmann constant, and A is the surface area of the body

$$\text{Therefore, for a sphere, } \frac{dQ}{dt} \propto r^2 T^4$$

$$\text{Since we are given that } \left(\frac{dQ}{dt}\right)_A = \left(\frac{dQ}{dt}\right)_{SUN},$$

$$r_A^2 T_A^4 = r_{SUN}^2 T_{SUN}^4$$

$$\Rightarrow \left(\frac{T_A}{T_{SUN}}\right)^4 = \left(\frac{r_{SUN}}{r_A}\right)^2$$

$$\Rightarrow \frac{T_A}{T_{SUN}} = \left(\frac{r_{SUN}}{r_A}\right)^{1/2} = \frac{1}{2} \Rightarrow T_A = \frac{1}{2} T_{SUN} = 3000 \text{ K}$$

- 5.(B) Work = Area under P-V curve

$$= \frac{1}{2}(3P_0 - P_0)(3V_0 - V_0) + P_0(3V_0 - V_0) = 4P_0V_0$$

- 6.(A) Restoring force, $F = -(kx + 2kx) = -(3k)x$

$$\text{Therefore, the time period, } T = 2\pi\sqrt{\frac{m}{3k}}$$

- 7.(C)** Let the area of cross-section of the blocks be A_0

Let the temperature at the interface be T

Since the rate of heat flow through the blocks must be equal, $\frac{KA_0(T_A - T)}{L} = \frac{(3K)A_0(T - T_B)}{2L}$

Solving, we get $T = \frac{1}{5}(2T_A + 3T_B)$

- 8.(D)** Tension in wire 1, $T_1 = (m_A + m_B)g = 6g$

Tension in wire 2, $T_2 = m_B g = 2g$

We know that $\text{Strain} = \frac{\text{Stress}}{\text{Young's modulus}} = \frac{T}{AY}$

Therefore, $\frac{\text{Strain}_1}{\text{Strain}_2} = \left(\frac{T_1}{A_1 Y_1}\right) \left(\frac{A_2 Y_2}{T_2}\right) = \left(\frac{T_1}{T_2}\right) \left(\frac{A_2}{A_1}\right) \left(\frac{Y_2}{Y_1}\right) = \left(\frac{6g}{2g}\right) \left(\frac{2}{1}\right) \left(\frac{1}{3}\right) = 4$

- 9.(B)** We know that for linear SHM, $KE_{\max} = \frac{1}{2} m \omega^2 A^2$

Therefore, analogously for angular SHM, $KE_{\max} = \frac{1}{2} I \omega^2 \theta_m^2 = \frac{1}{2} (mL^2) \left(\frac{g}{L}\right) \theta_m^2 = \frac{1}{2} mgL \theta_m^2$

- 10.(A)** Restoring torque (about O) acting on the rod, $\tau = mg \left(\frac{L}{4} \sin \theta\right) \approx \left(\frac{mgL}{4}\right) \theta$

Moment of inertia of the rod about O , $I = \frac{1}{12} mL^2 + m \left(\frac{L}{4}\right)^2 = \frac{7}{48} mL^2$

We know that if restoring torque is $\tau = C\theta$, the time period of angular SHM is $2\pi \sqrt{\frac{I}{C}}$

Therefore, the time period of angular SHM of the rod, $T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{\left(\frac{7}{48} mL^2\right)}{\left(\frac{mgL}{4}\right)}} = 2\pi \sqrt{\frac{7L}{12g}}$

- 11.(C)** We know that $\text{Specific heat} = \frac{\text{Molar specific heat}}{\text{Molar mass}}$

Therefore, the specific heat at constant volume of Argon gas (which is a monoatomic gas) is

$$C = \frac{\left(\frac{3R}{2}\right)}{0.04} = \frac{75R}{2} = 311.63 \text{ J/kg}$$

- 12.(D)** Since the block is initially at rest, the initial position is the upper extreme

The mean position is where the block is in equilibrium. Therefore, at the mean position, the

elongation in the spring is $\frac{mg}{k} = \frac{(0.3)(10)}{120} = 2.5 \text{ cm}$

Since initially the spring was in its natural length, the amplitude of the SHM is 2.5 cm

Now, the angular frequency of the SHM, $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{120}{0.3}} = 20 \text{ rad/s}$

So, the maximum velocity of the block during the SHM,

$$v_{\max} = A\omega = (0.025)(20) = 0.5 \text{ m/s} = 50 \text{ cm/s}$$

Since the initial velocity is zero, and the velocity is negative for the next half time period before becoming zero again at the lower extreme position, the variation with time is $v(t) = -v_{\max} \sin(\omega t)$
[This can be visualised readily using the graph of the sine function]

Therefore, $v(t) = -50 \sin(20t) = 50 \cos\left(20t + \frac{\pi}{2}\right)$

13.(C) Let the final temperature be T_f

Heat gained by ice = Heat lost by water

$$m_1 L + m_1 C(T_f - 0) = m_2 C(T - T_f) \Rightarrow T_f = \frac{m_2 T - m_1 \left(\frac{L}{C}\right)}{(m_1 + m_2)}$$

14.(C) Work done by the gas is positive if and only if the gas expands, regardless of the type of the process

For adiabatic expansion, $\Delta Q = 0$ and $W > 0$

Therefore, $\Delta U < 0$

For isobaric expansion, $W > 0$

And, $\Delta U = nC_v \Delta T = \frac{C_v}{R} P \Delta V > 0$

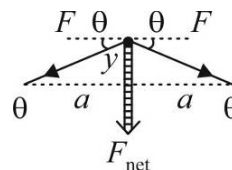
15.(A) Net restoring force,

$$F_{\text{net}} = 2F \sin \theta \Rightarrow F_{\text{net}} = 2 \left(\frac{GMm}{a^2 + y^2} \right) \left(\frac{y}{\sqrt{a^2 + y^2}} \right) \Rightarrow F_{\text{net}} = \frac{2GMmy}{(a^2 + y^2)^{3/2}}$$

Since $y \ll a$, $F_{\text{net}} = \left(\frac{2GMm}{a^3} \right) y$

We know that if $F_{\text{restoring}} = ky$, the frequency of the SHM is $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$

Therefore, the frequency, $f = \frac{1}{2\pi} \sqrt{\frac{2GM}{a^3}}$



16.(C) Considering a mass m of the metal, $\left(\frac{m}{\rho_2}\right) = \left(\frac{m}{\rho_1}\right)(1 + \gamma(T_2 - T_1)) \Rightarrow \gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1\right)$

17.(B) Since $\frac{P_1}{V_1^3} = \frac{P_2}{V_2^3} \Rightarrow P_2 = \left(\frac{V_2}{V_1}\right)^3 P_1$

For $PV^{-3} = \text{constant}$, $C = C_v - \frac{R}{-3-1} = \frac{5}{2}R + \frac{R}{4} = \frac{11R}{4}$

$$\therefore \Delta Q = nC\Delta T = \frac{11}{4}nR(T_2 - T_1) = \frac{11}{4}(P_2V_2 - P_1V_1) = \frac{11}{4} \left(\frac{V_2^3}{V_1^3} P_1V_2 - P_1V_1 \right) = \frac{11}{4} \frac{P}{V_1^3} (V_2^4 - V_1^4)$$

18.(D) Bulk modulus is defined as: $B = -\left(\frac{\Delta P}{\frac{\Delta V}{V}}\right)$

Therefore, the change in volume due to the difference in pressure, $(\Delta V)_{\text{Pressure}} = -\left(\frac{\Delta P}{B}\right)V$

And, the change in volume due to the difference in temperature, $(\Delta V)_{\text{Thermal}} = V\gamma\Delta T$

Therefore, the total change in volume, $(\Delta V)_{\text{Total}} = (\Delta V)_{\text{Pressure}} + (\Delta V)_{\text{Thermal}} = V\left(-\frac{\Delta P}{B} + \gamma\Delta T\right)$

So, the fractional change in volume,

$$f = \frac{(\Delta V)_{\text{Total}}}{V} = \left(-\frac{\Delta P}{B} + \gamma \Delta T \right) = \left(\frac{-(10^3)(10)(300)}{150 \times 10^9} + (3 \times 10^{-6})(10 - 20) \right) = -(5 \times 10^{-5})$$

- 19.(C)** Remember that according to Newton's law of cooling, the time taken by a body to cool from temperature T_1 to temperature T_2 in a room at temperature T_0 is proportional to $\log_e \left(\frac{T_1 - T_0}{T_2 - T_0} \right)$

But when T_1 and T_2 are very close together (or more properly, $T_1 - T_2 \ll T_1$), the rate of cooling (i.e. $\frac{dT}{dt}$) can be considered constant for the brief time it takes the body to cool from temperature T_1 to T_2 . And since we know that according to Newton's law of cooling, the rate of cooling is proportional to $(T - T_0)$, the rate of cooling during cooling from T_1 to T_2 is very nearly proportional to $(T_1 - T_0)$ [or $(T_2 - T_0)$, since T_1 and T_2 are very close together].

And, it is intuitive that if the rate of cooling is constant, the time taken to cool from T_1 to T_2 is

$$t_{1 \rightarrow 2} = \frac{T_1 - T_2}{\text{The constant rate of cooling}}$$

Therefore,
$$t_{1 \rightarrow 2} \propto \left(\frac{T_1 - T_2}{T_1 - T_0} \right)$$

So, the time taken to cool from 80.0°C to 79.9°C ,
$$t_1 \propto \left(\frac{80.0 - 79.9}{80.0 - 20.0} \right)$$

And, the time taken to cool from 40.0°C to 39.9°C ,
$$t_2 \propto \left(\frac{40.0 - 39.9}{40.0 - 20.0} \right)$$

Therefore,
$$\frac{t_1}{t_2} = \frac{1}{3}$$

- 20.(D)** We know that the heat absorbed by the gas in an isobaric process, $\Delta Q = nC_p \Delta T = nC_p (T_f - T_i)$

Also, we know that the rms speed of the molecules at temperature T ,
$$v_{\text{RMS}} = \sqrt{\frac{3RT}{M_0}}$$

Here, M_0 is the mass of one mole of the gas

Therefore,
$$T = \frac{M_0}{3R} v_{\text{RMS}}^2$$

So,
$$\Delta Q = \frac{nM_0 C_p}{3R} (v_{\text{RMS},f}^2 - v_{\text{RMS},i}^2) = \frac{7m}{6} (v_{\text{RMS},f}^2 - v_{\text{RMS},i}^2)$$

[because nM_0 is simply the total mass of the gas, m , and for a diatomic gas, $C_p = \frac{7R}{2}$]

Therefore,
$$\Delta Q = \left(\frac{7}{6} \right) (0.1) (700^2 - 500^2) = 28 \text{ kJ}$$

SECTION-2

- 21.(30) Let the specific heat of the liquids be S_X and S_Y

Then, for the first mixing,

Heat lost by liquid X = Heat gained by liquid Y

$$\Rightarrow 10S_X(80 - 32) = 20S_Y(32 - 20) \Rightarrow 2S_X = S_Y$$

Now, let the final temperature after the second mixing be T

So, for the second mixing,

Heat lost by the mixture = Heat gained by liquid Y

$$\Rightarrow (10S_X + 20S_Y)(32 - T) = 5S_Y(T - 20)$$

Replacing $S_Y = 2S_X$ and solving, we get $T = 30^\circ\text{C}$

- 22.(8) If the heat absorbed and the heat rejected per cycle are Q_H and Q_L , we know that

$$\frac{Q_H}{Q_L} = \frac{T_H}{T_L} = \frac{500}{300} = \frac{5}{3}$$

We are given that $Q_H = 20 \text{ kJ} \Rightarrow Q_L = 12 \text{ kJ}$

By the first law of thermodynamics (energy conservation), the work done by the gas per cycle,

$$W = Q_H - Q_L = 20 - 12 = 8 \text{ kJ}$$

- 23.(84) We can use the result for speed as a function of position,

$$v(y) = \omega\sqrt{A^2 - y^2}$$

Therefore, $\omega\sqrt{A^2 - (-1)^2} = 24$ and $\omega\sqrt{A^2 - 5^2} = 12\sqrt{2}$

Dividing the equations and squaring,

$$\frac{A^2 - 1}{A^2 - 25} = 2 \Rightarrow A = 7 \text{ cm}$$

Therefore, $\omega = \frac{24}{\sqrt{7^2 - 1}} = \sqrt{12} \text{ rad/s}$

So, the maximum acceleration, $a_{\max} = \omega^2 A = (12)(7) = 84 \text{ cm/s}^2$

- 24.(21.50) $\frac{dQ}{dt} = \frac{KA(T_1 - T_2)}{L} \Rightarrow 1800 = \frac{(1)(1.2)(T_1 - 14)}{0.5 \times 10^{-2}} \Rightarrow T_1 = 21.5^\circ\text{C}$

- 25.(62.50)

Let the number of moles of each gas be n and let the temperature be T

Total KE of Nitrogen molecules, $K_1 = \frac{5}{2}nRT$

Total KE of Helium molecules, $K_2 = \frac{3}{2}nRT$

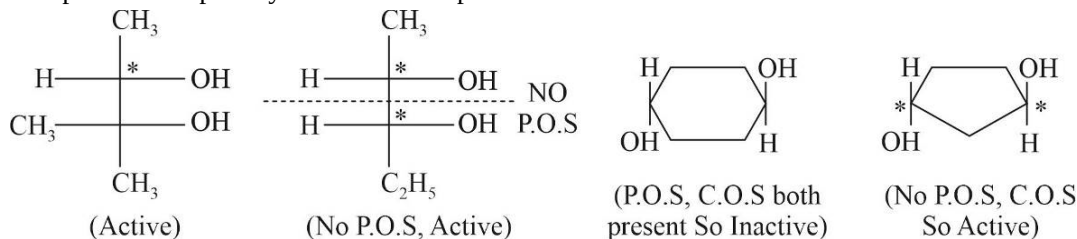
Therefore, the required percentage $= \frac{K_1}{K_1 + K_2} \times 100 = 62.5$

Chemistry

SECTION-1

- 1.(D) Compound D is aromatic as it obey Huckel rule of cyclic conjugation of $(4n + 2)\pi e^-$
($n = 0, 1, 2, \dots$)

- 2.(C) Compound 3 is optically inactive due to presence of P.O.S and C.O.S

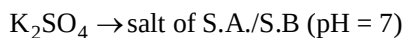


3.(B) $K_{a1} \times K_{a2} = \frac{[H^+]^2 (S^{2-})}{[H_2S]}$

$$10^{-20} = \frac{[10^{-1}]^2 (S^{2-})}{0.1}$$

$$S^{2-} = \frac{10^{-20} \times 0.1}{10^{-2}} = 10^{-19}$$

- 4.(B) $NH_4Cl \rightarrow$ salt of S.A/W.B ($pH < 7$)



- 5.(A) It is a disproportionation redox reaction

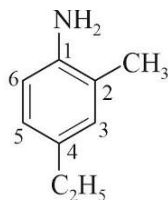


$$\therefore \text{Equivalent mass of } Z_2 = \frac{M^o}{5/3} = \frac{3}{5}[160] = 96$$

- 6.(B) $AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2Cl$

$$K = K_{sp} \times K_f = 10^{-4}$$

- 7.(A)



(4-Ethyl-2-methyl aniline)

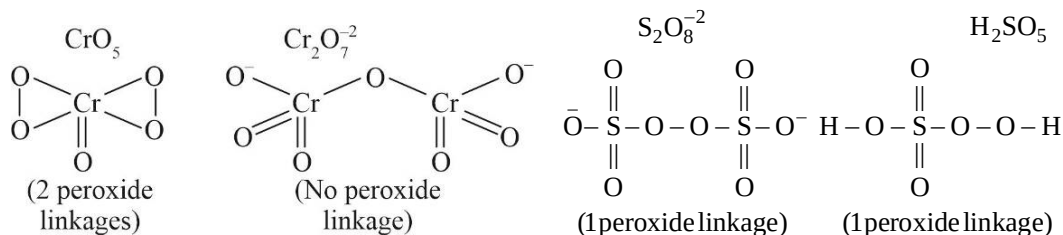
- 8.(C) Acidic buffer having $pH = pK_a$ has maximum buffer capacity.

- 9.(C) As at half neutralization point

$$pOH = 14 - pH = 14 - 8 = 6$$

$$\Rightarrow pK_b = 6 \dots \dots \text{ for } AgOH$$

10.(A)



11.(B) As for a H_2O_2 solution volume strength = $N \times 5.6$

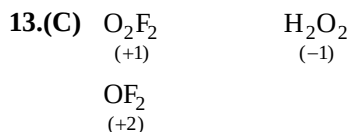
$$\Rightarrow N = \frac{2.8}{5.6} = \frac{1}{2} = 0.5$$

12.(B) KMnO_4 will oxidise Fe^{+2} into Fe^{+3} (in the acidic medium)

$$\text{meq of } \text{KMnO}_4 = \text{meq of } \text{Fe}^{2+}$$

$$M \times 5 \times 0.01 = (0.01 \times 1)$$

$$\text{Molarity of } \text{KMnO}_4 = \frac{1}{5} = 0.20 \text{ M}$$



$\therefore$ increasing order of oxidation of the oxygen atoms is option (C).

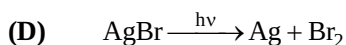
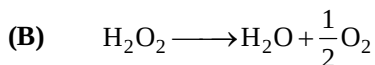
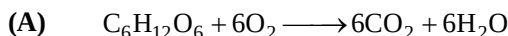
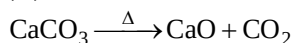
14.(B) On adding HCl , pH of the solution will decrease slowly initially; and sharply near the equivalent point.

$$\text{pH} = 7 + \frac{1}{2} [\text{pK}_a - \text{pK}_b]$$

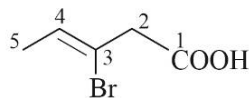
$(\text{CH}_3\text{COOH}) \quad (\text{NH}_4\text{OH})$

$$\therefore \text{pH} = 7 \text{ [as } \text{pK}_a = \text{pK}_b = 5 \text{]}$$

16.(C) (C) is not redox reaction

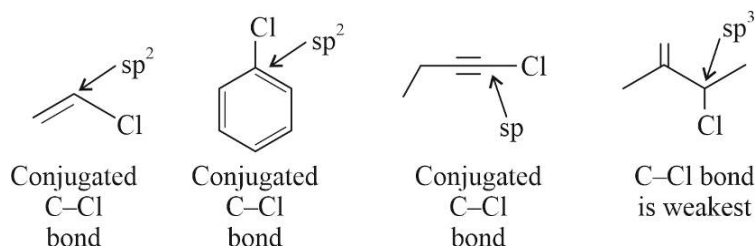


17.(A)

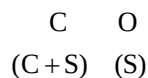
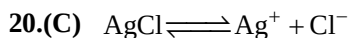


3-Bromo-pent-3-en-oic acid

18.(D)



- 19.(B) Eclipsed conformer of 1, 2-dichloro ethane is less stable than its anti-conformer; twisted boat form of cyclohexane is more stable than its Boat form. Chair form of cyclohexane is optically inactive; Gauche conformer of ethane-1, 2-diol is more stable due to intra H-bonding.



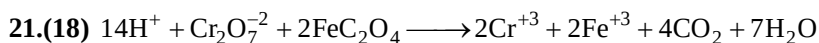
$$K_{sp} = (\text{C} + \text{S})(\text{S})$$

Neglecting "S" compared to "C"

$$K_{sp} = \text{C}(\text{S})$$

$$\text{S} = \frac{10^{-10}}{10^{-1}} = 10^{-9} \text{ M}$$

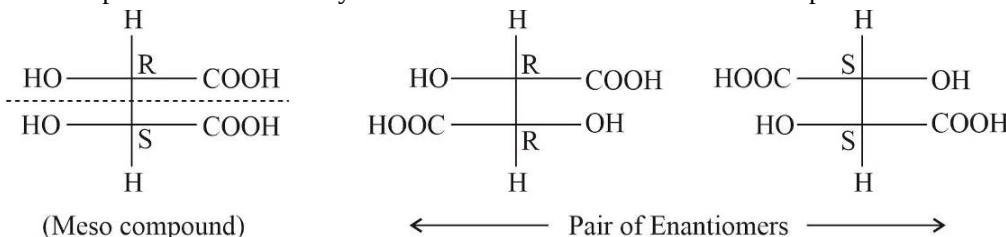
SECTION-2



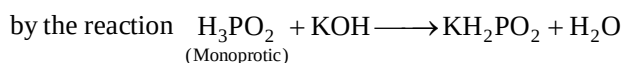
$$\therefore n_1 = 1, n_2 = 2, n_3 = 2, n_4 = 2, n_5 = 4, n_6 = 7$$

$$\text{And } (n_1 + n_2 + n_3 + n_4 + n_5 + n_6) = 18$$

- 22.(3) The compound consists of only 2 identical chiral centres thus number of possible stereoisomers is 3.



- 23.(7.50) At end point we will obtain an aqueous solution of KH_2PO_2



$$\text{Here } [\text{KH}_2\text{PO}_2] = \frac{10 \times 1/5}{(10 + 10)} = \frac{1}{10} = 0.1$$

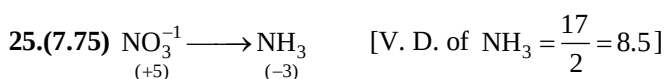
$$\text{pH} = 7 + \frac{1}{2}[\text{p}K_a + \log[\text{H}_2\text{PO}_2^-]_{\text{salt}}] = 7 + \frac{1}{2}[2 - 1] = 7 + 0.5 = 7.5$$

- 24.(6) Least soluble metal chloride out of the given two is Hg_2Cl_2 ($K_{sp} = 10^{-16}$)

To just start precipitation of Hg_2Cl_2 let concentration of HCl required is "C", such that

$$10^{-16} = 10^{-4}[\text{C}]^2$$

$$(\text{or}) \quad \text{C}^2 = \frac{10^{-16}}{10^{-4}} = 10^{-12} \quad (\text{or}) \quad \text{C} = 10^{-6} \text{ M} = [\text{H}^+] \quad \therefore \quad \text{pH of the solution} = 6$$



Hence for NO_3^- [$M^\circ = 62$]

$$\text{Equivalent mass} = \frac{62}{8} = 7.75$$

Mathematics

SECTION-1

1.(B) $2^{(2\cos^2 x - 1)} = 3 \cdot 2^{\cos^2 x} - 4$

$2^{\cos^2 x} = t$

$\frac{t^2}{2} = 3 \cdot t - 4 \Rightarrow t^2 - 6t + 8 = 0 \quad t = 2, t = 4$

$2^{\cos^2 x} = 2^1 \text{ or } 2^2$

$\cos^2 x = 1 \quad \text{or} \quad 2 \text{ (not possible)}$

$x = n\pi \quad \pi, 2\pi, 3\pi, \dots, 100\pi$

Sum = 5050π

2.(B) $S_{10} = \frac{10}{2}(2a + 9d) = 0 \Rightarrow d = \frac{-2a}{9}$

$S_{20} = S_{10} + \text{Next 10 terms sum}$

$= -200$

$+\frac{20}{2}(2a + 19d) = -200$

$2a + 19\left(-\frac{2a}{9}\right) = -20$

$-\frac{20a}{9} = -20 \Rightarrow a = 9$

3.(A) $\tan \theta + \cot \theta = (\tan \theta + \cot \theta)^2 - 2$

$\tan \theta + \cot \theta = t$

$t^2 - t - 2 = 0$

$(t - 2)(t + 1) = 0 \Rightarrow t = 2 \text{ or } (t = -1) \text{ Rejected as } \theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\tan \theta + \cot \theta = 2 \Rightarrow \tan \theta = 1$

$\sin \theta = \frac{1}{\sqrt{2}}, \cos \theta = \frac{1}{\sqrt{2}}$

4.(B) $T_r = r^2 \cdot \frac{{}^{26}C_r}{{}^{26}C_{r-1}} = \frac{r^2(27-r)}{r} = r(27-r)$

$T_r = 27r - r^2$

$S = \sum_{r=1}^{10} 27r - r^2 = \frac{27 \times 10 \times 11}{2} - \frac{10 \times 11 \times 21}{6} = \frac{10 \times 11}{2} [27 - 7] = 55 \times 20 = 1100$

5.(A)

$\begin{array}{cccccc} \square & \square & \square & \square & \square & \square \\ \downarrow & \downarrow & & \downarrow & & \\ 4 & 4 & 3 & 3 & 1 & 2 \end{array}$

$16 \times 9 \times 10 = 1440$

$\begin{array}{cc} 1 & 6 \\ 3 & 2 \\ 3 & 6 \\ 5 & 2 \\ 5 & 6 \\ 7 & 2 \\ 7 & 6 \\ 9 & 2 \\ 9 & 6 \end{array}$

6.(A) $y^2 = 8x$

$P(4, 6) \quad y = mx + \frac{2}{m}$

$6 = 4m + \frac{2}{m} \Rightarrow 4m^2 - 6m + 2 = 0$

$\Rightarrow 2m^2 - 3m + 1 = 0$

$2m^2 - 2m - m + 1 = 0$

$2m(m-1) - 1(m-1) = 0$

$m = 1, m = \frac{1}{2}$

$y = x + 2 \text{ or } y = \frac{x}{2} + 4 \Rightarrow 2y = x + 8$

7.(C) $(2^{1/2} + 3^{1/5})^{10}$

$T_{r+1} = {}^{10}C_r (2^{1/2})^{10-r} (3^{1/5})^r$

For rational terms

$r = 0, r = 10$

$T_1 = 32, T_{11} = 9$

Sum = 41

8.(C) $\frac{x^2}{9} + \frac{y^2}{4} = 1$

$4x - 2y - 5 = 0 \quad (m = 2)$

Slope of Normal = 2

$\frac{3x}{\cos \theta} - \frac{2y}{\sin \theta} = 5$

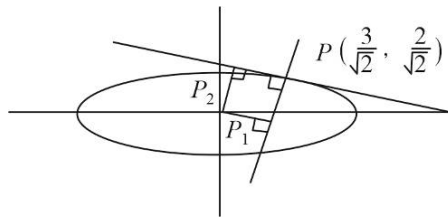
Slope = $\frac{3 \sin \theta}{2 \cos \theta} = 2 \Rightarrow \tan \theta = \frac{4}{3} \begin{cases} \sin \theta = \frac{4}{5} \\ \cos \theta = \frac{3}{5} \end{cases}$

9.(A) Tangent $\frac{x}{3\sqrt{2}} + \frac{y}{2\sqrt{2}} = 1$

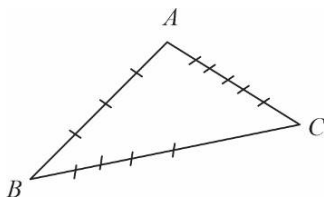
Normal $3x\sqrt{2} - 2y\sqrt{2} = 5$

$P_1 = \frac{5}{\sqrt{18+8}}, P_2 = \frac{1}{\sqrt{\frac{1}{18} + \frac{1}{8}}} = \frac{12}{\sqrt{26}}$

Area = $P_1 P_2 = \frac{60}{26} = \frac{30}{13}$



10.(D) Total number of triangle = $\left({}^{12}C_3 - \left({}^3C_3 + {}^4C_3 + {}^5C_3 \right) \right)$



$= 220 - (1 + 4 + 10) = 205$

11.(D) $x^2 = 8y \quad \frac{x^2}{3} + y^2 = 1$

$$\frac{8y}{3} + y^2 = 1$$

$$3y^2 + 8y - 3 = 0$$

$$y + 3 = 0 \text{ (Rejected)}$$

$$3y - 1 = 0$$

12.(C) $e_1 = \sqrt{1 - \frac{2}{3}} = \frac{1}{\sqrt{3}}$

$$e_1 e_2 = \frac{1}{2}$$

$$e_2 = \frac{\sqrt{3}}{2} \Rightarrow \frac{\sqrt{3}}{2} = \sqrt{1 - \frac{b^2}{16}} \Rightarrow 1 - \frac{b^2}{16} = \frac{3}{4} \Rightarrow b^2 = 4$$

Length of minor axis of $E_2 = 4$

13.(C) $a = 25 \quad d = \frac{-3}{5}$

For maximum sum $T_n > 0$

$$25 + (n-1)\left(\frac{-3}{5}\right) > 0 \quad \frac{(n-1)3}{5} < 25 \Rightarrow n-1 < \frac{125}{3} \Rightarrow n < \frac{128}{3} \quad n < 42.66$$

14.(D) $T_{r+1} = {}^{10}C_r (x \sin \theta)^{10-r} \left(\frac{\cos \theta}{x}\right)^r$

$$= {}^{10}C_r (\sin \theta)^{10-r} (\cos \theta)^r x^{10-2r}$$

$$10 - 2r = 0 \Rightarrow r = 5$$

$$T_6 = {}^{10}C_5 (\sin \theta)^5 (\cos \theta)^5 = \frac{{}^{10}C_5}{2^5} (\sin 2\theta)$$

$$(T_6)_{\max} = \frac{{}^{10}C_5}{2^5}$$

15.(B) C A R A, O, E, C, S, S, N, N

$$\frac{6!}{2!2!} \times {}^7C_3 \times 3! = \frac{6!}{2!2!} \times \frac{7!}{4!3!} \times 3!$$

$$= \frac{6 \times 5}{4} \times 7 \times 6 \times 5! = 15 \times 21 \times 5!$$

16.(A) $S = 1 + \left(1 + \frac{1}{5}\right)\left(\frac{1}{2}\right) + \left(1 + \frac{1}{5} + \frac{1}{5^2}\right)\left(\frac{1}{2^2}\right) + \dots$

$$\frac{S}{2} = \frac{\left(\frac{1}{2}\right)}{\left(1 + \frac{1}{5}\right)} + \frac{\left(1 + \frac{1}{5}\right) \times \frac{1}{2^2}}{\left(1 + \frac{1}{5} + \frac{1}{5^2}\right)} + \dots$$

$$\frac{S}{2} = 1 + \frac{1}{5} \times \frac{1}{2} + \frac{1}{5^2 \times 2^2} + \frac{1}{5^3} \times \frac{1}{2^3} + \dots$$

$$\frac{S}{2} = \frac{1}{1 - \frac{1}{10}} \Rightarrow \frac{S}{2} = \frac{10}{9} \Rightarrow S = \frac{20}{9}$$

$$17.(B) \quad E = \left(a^4 + 4 + \frac{1}{a^4} \right) \left(b^2 + \frac{1}{b^2} + 3 \right) ((c+1)^2 + 1)$$

$$E_{\min} = 6 \times 5 = 30$$

$$18.(B) \quad \frac{(\sin 7\theta + \sin 5\theta) + 5(\sin 5\theta + 5\sin 3\theta) + 12(\sin 3\theta + \sin \theta)}{\sin 6\theta + 5\sin 4\theta + 12\sin 2\theta}$$

$$\frac{2\sin 6\theta \cos \theta + 5 \times 2\sin 4\theta \cos \theta + 12 \times 2\sin 2\theta \cdot \cos \theta}{\sin 6\theta + 5\sin 4\theta + 12\sin 2\theta}$$

$$= 2\cos \theta = 2\cos\left(\frac{\pi}{5}\right) = 2\sin\left(\frac{3\pi}{10}\right)$$

$$19.(B) \quad (y-2)^2 = 16(x-1)$$

Focus (5, 2)

$$\text{Focal chord} \Rightarrow (y-2) = m(x-5)$$

$$\text{Tangent to } x^2 + y^2 - 14x - 4y + 51 = 0$$

$$\text{Centre (7, 2) } r = \sqrt{2}$$

$$\left| \frac{2m}{\sqrt{1+m^2}} \right| = \sqrt{2} \Rightarrow 4m^2 = 2m^2 + 2$$

$$2m^2 = 2$$

$$m = 1 \quad \text{or} \quad -1$$

$$20.(D) \quad \text{Distance b/w L.R. and TAV} = \frac{4}{\sqrt{2}}$$

$$\Rightarrow \text{L. R.} = 4 \times \frac{4}{\sqrt{2}} = \frac{16}{\sqrt{2}} = 8\sqrt{2}$$

SECTION-2

$$21.(3) \quad T_{r+1} = {}^nC_r (\sqrt{x})^{n-r} \frac{1}{2^r \cdot (x^{1/4})^r} = {}^nC_r \frac{1}{2^r} \times x \left(\frac{n-r}{2} - \frac{r}{4} \right)$$

Coefficient of first three terms are in A.P.

$${}^nC_0 \frac{1}{2^0}, {}^nC_1 \frac{1}{2}, {}^nC_2 \frac{1}{2^2} \rightarrow \text{A.P.}$$

$${}^nC_1 = 1 + \frac{{}^nC_2}{4}$$

$$4n = 4 + \frac{n(n-1)}{2} \Rightarrow n^2 - 9n + 8 = 0 \Rightarrow n = 8 \text{ or } n = 1. \text{ (Not possible)}$$

$$\text{Exponent of } x = \frac{16-3r}{4} \quad r = 0, 4, 8$$

$$22.(4) \quad T_r = \frac{6r+1}{(3r-1)^2(3r+2)^2} = \frac{6r+1}{(9r^2-6r+1)(9r^2+12r+4)}$$

$$= \frac{1}{3} \left(\frac{(9r^2+12r+4) - (9r^2-6r+1)}{(9r^2+12r+4)(9r^2-6r+1)} \right)$$

$$T_r = \frac{1}{3} \left(\frac{1}{(3r-1)^2} - \frac{1}{(3r+2)^2} \right)$$

$$T_1 = \frac{1}{3} \left[\frac{1}{2^2} - \frac{1}{5^2} \right]$$

$$T_2 = \frac{1}{3} \left[\frac{1}{5^2} - \frac{1}{8^2} \right]$$

$\vdots$

$$T_{10} = \frac{1}{3} \left(\frac{1}{(29)^2} - \frac{1}{(32)^2} \right)$$

$$\text{Sum} = \frac{1}{3} \left[\frac{1}{2^2} - \frac{1}{(32)^2} \right] \quad \text{Sum} = \frac{85}{1024}$$

23.(32) $y = \frac{x}{\sqrt{3}} + 12\sqrt{3}$ (Tangent)

$$y = \frac{x}{\sqrt{3}} - \frac{24}{\sqrt{3}} - \frac{12}{3\sqrt{3}} \quad (\text{Normal})$$

$$\text{Distance} = \frac{20\sqrt{3} + \frac{4}{\sqrt{3}}}{\sqrt{1 + \frac{1}{3}}} = 32$$

24.(5400) ${}^5C_2 \times {}^4C_2 \times {}^6C_2 \times {}^4C_2$
 $10 \times 6 \times 15 \times 6 = 5400$

25.(10) $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$; $\frac{a}{\cos \theta 4a} = \frac{-b}{\sin \cdot 3b} = \frac{a^2 - b^2}{24}$

$$\tan \theta = -\frac{4}{3}, a^2 - b^2 = -\frac{24}{3 \times \left(-\frac{4}{5}\right)}$$

$$a^2 - b^2 = 10$$